

# ITAMAR ROCHA FILHO

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**Languages** Portuguese, English, French, Spanish.

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## EDUCATION

**Harvard University** — M.Eng., Computational Science and Engineering Expected 05/2026  
· Teaching Fellow, AC215 (Machine Learning Operations)  
· Research: RLVR / LLM post-training (advisor: Sham Kakade)

**B.S. in Computer Engineering - Universidade Federal da Paraíba (Brazil) - 9.59/10 (1st)** 07/2018 – 05/2024

## RESEARCH & PUBLICATIONS

**Parameter-Efficient Reinforcement Learning Using Prefix Optimization** Under review (ICLR 2026)  
· Co-first author. Proposed *Prefix-RL*: train a small adapter to generate the first  $k$  tokens, steering a frozen target model (inference-only).  
· Improved Qwen-7B Math-500 pass@1 from 67.4%→74.4% (+7.0) with gains on AIME/AMC23/Minerva; demonstrated physics transfer and steering of FP8 quantized targets.  
· 300–3000× lower RL training compute (e.g., ~3,000× for Qwen2.5-1.5B→72B) and reduced practical GPU needs (70B tuned with 8 GPUs).

**Predicting RL Fine-Tuning Success (Master’s thesis)** Ongoing  
· Curated ~65B tokens of math/reasoning data; trained ~40 OLMo models (150M/1B) from scratch and ran PPO on GSM8K & MATH to collect RL outcomes.  
· Building a reusable library of predictive metrics (perplexity, coverage/diversity, gradient-based, pivotal-token) to study empirical “scaling laws” for RL data/model selection.

**Evolutionary Alignment** Manuscript in preparation (targeting ICML 2026)  
· Replicated/extended Qiu et al. (2025): ES (Evolution Strategies) can avoid reward hacking by GRPO on Conciseness, but only in a narrow  $(\alpha, \sigma)$  region.  
· Analyzed ES vs GRPO solution geometry (perturbations, linear mode connectivity, cosine similarity); showed ES achieves helpful–harmless Pareto improvements on PKU-SafeRLHF using only 250 examples (approximate 1% of the data used in the original Safe-RLHF paper) and no gradient access.  
· [In progress] Created checkpoint storage optimization, saving storage by loading and reproducing the checkpoints from noise + rewards (Reduced storage from 12GB to 1.1MB for ES checkpoints).

**Evaluating Misalignment Under Pressure** CS2881r project (Nov 2025)  
· Built an agentic coding evaluation with an “impossibility knob” and 24 pressure scenarios (threats × prohibitions × difficulty).  
· Found agents often hack evaluations when honest solutions are impossible (sometimes despite explicit prohibition), with strong cross-model differences; piloted LLM-based monitoring.

**An Approach for Automatic Description of Characters for Blind People** WebMedia 2021

**Iris-CV: Classifying Iris Flowers Is Not as Easy as You Thought** BRACIS 2021

## EXPERIENCE

**Researcher - Kempner Institute (Harvard)** 05/2025 – Present  
· LLM post-training research on RLVR/RLHF and predictive metrics for RL fine-tuning success (Master’s thesis).  
· Built end-to-end multi-node RLHF on Qwen-7B (Ray + distributed PPO via OpenRLHF); resolved NCCL hangs/sync/OOM issues and added fault-tolerant job arrays.  
· Led OOD physics evaluation and LLM-as-judge pipeline for Prefix-RL; enabled large experimental sweeps.

**SWE/MLE - about.webdraw.com (Joined before the seed round)** 11/2023 - 06/2024  
I developed the company’s first AI product: a LLM-based Sales Assistant. Also worked on infrastructure, created a tiered cache system, created the internal load test, sponsored competitions, and hired new individuals. I also created an AI gateway that integrated 20+ different generative image, audio, text, and video providers in a uniform pattern and made a hands-free mode leveraging voice models.

**Software Engineer Intern / Research - Google [Mountain View - CA]** 06/2023 - 09/2023  
I interned at the YouTube Strategic Video Compression team, working on an open-ended research project. The project involved taking different approaches to improve the ultra-popular video compression algorithm. I cut the cost of the component by 50% while also decreasing the model’s error by 7% and simplifying the code base to one model instead of six.

Software Engineer Intern - Meta [London - UK]01/2023 - 04/2023

I worked on the App Ads Platform team. I was responsible for creating a new feature for Instagram Ads. The project consisted of creating an end-to-end demo (Back end + Front end: web and mobile), testing, Analyzing metrics, Resolving bugs, and collaborating with other teams. I completed the project in advance and started working on side projects to help the team.

Software Engineer Machine Learning Intern - Meta [London - UK]01/2022 - 04/2022

I worked on the AI for AR team. The team is focused on cutting-edge research in computer vision and audio. My internship involved adapting a state-of-the-art model to use Meta’s production environment, cleaning, refactoring the code, and deploying it for further testing and production use. I also built a demo for internal use.

Software Engineering Intern - Google [Belo Horizonte - BR]08/2021 - 11/2021

I worked on the Google Search - contextualization team. My work involved exploring and analyzing a new data source creating an end-to-end demo (Data Generation + Analysis + Back-end + Front-end). After completing my demo, I increased the original feature’s coverage by 68.38% while maintaining a high data quality.

SKILLS

|             |                                                                            |
|-------------|----------------------------------------------------------------------------|
| Programming | Python, C/C++, TypeScript, Julia.                                          |
| ML/LLM      | PyTorch, TensorFlow, vLLM, OpenRLHF, verl, RLVR/RLHF, TRL.                 |
| Systems     | Ray, Git, Linux, Kubernetes, GCP, AWS.                                     |
| Interests   | Post-training, algorithms/optimization, distributed training/infra, MLOps. |

EXTRACURRICULAR ACTIVITIES

Ex-President, Co-Founder and Advisor TAIL - UFPB08/2020 - 12/2025

Technology and Artificial Intelligence League is a non-profit organization aimed at teaching, producing, contributing to research, and helping members get started on internships, graduation programs, and full-time jobs. It currently has 65 active members and has introduced hundreds of people from multidisciplinary areas to programming and AI.

Founder and Advisor - Momento08/2024 - Present

Momento is an NGO that focuses on mapping alumni from public universities in Brazil and generating value through mentorship, events, and competitions. We are currently running the second mentorship cohort, with more than 200 student applications from all over Brazil, and mentors from industry and academia, including representatives from MIT, Harvard, DeepMind, Google, Cohere, and many others.

HONORS AND AWARDS

NXTP Fellow11/2025

Invited to be part of the NXTP Fellowship network. Receiving a monthly stipend, mentorship, and access to the firm’s resources.

AGI House Fellow06/2025

Selected to spend the summer at the AGI House in California, working on my projects.

Líder Estudiar - Estudiar Foundation - NGO09/2021 - 05/2026

The most competitive scholarship in Brazil. 0.04% acceptance rate. Paid for my master’s program.

Teaching Initiation Award - UFPB11/2019

Awarded for the work done in the tutoring of Boolean Algebra and Logic Circuits class.